- *Mission modularity*: Modular payload bays and open architectures permit rapid reconfiguration for different missions (ISR, EW, communications relay, strike, maritime patrol) without major airframe redesign.

4.3.2 Hardware components

A typical third-generation UAV is an integrated system. The major hardware elements include:

- *Airframe and aero structure*: Optimized for endurance and low observability where required. Materials include composites and RAM (radar-absorbent materials) for UCAV variants. Aerodynamic design balances loiter efficiency, payload carriage, and low-drag cruise.
- *Propulsion system*: Turboprops, small turbofans, or efficient piston/electric hybrid systems for MALE platforms; fuel-efficient engines tuned for long loiter times and restricted IR signature in some military designs.
- *Flight control and avionics suite (FCU)*: High-reliability flight computers (often multiple redundant units), digital autopilot, EFIS (electronic flight instrumentation) subsystems, and actuator control for surfaces and control effectors.
- *Inertial navigation and guidance (INS/IMU) + GNSS*: High-grade IMUs coupled with multi-constellation GNSS receivers (GPS, GLONASS, Galileo, BeiDou) and tightly-coupled GNSS/INS algorithms for precise navigation and robustness against signal degradation.
- *Communications payloads*: Multi-band radios (UHF/VHF, L-band, Ku/Ka SATCOM terminals), datalinks supporting both low-latency